AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM354
Software Engineering

Midterm Examination (MTA) - Makeup
Fall Semester 2025/2026

Date: 6 /Dec./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part I – Multiple Choice Questions — 3 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. A system which is easier to understand, modify, replace, and reuse should:
   a. be loosely coupled
   b. be highly coupled
   c. have circular dependency
   d. have chain dependency

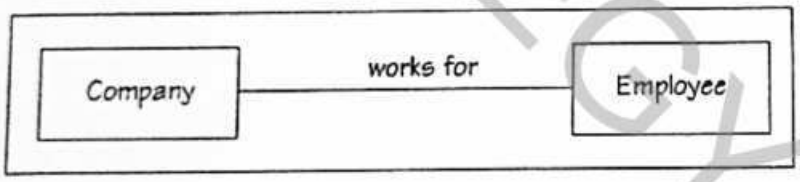
2. To which category and type do you classify the requirement:
   "The system must allow users to verify their accounts using their mobile number"
   a. Functional, Business
   b. Functional, Technical
   c. Non-functional, Operational
   d. Non-Functional, Cultural

3. Which of the following is not something that Use Case diagrams help provide:
   a. Eliciting requirements
   b. Representing requirements
   c. Identifying components
   d. Validating software systems

Part II – Short Answer Questions — 12 Marks

*Answer **all** the following short answer questions. Each question is worth 3 Marks.*

1. A new generative AI needs to be developed. State 3 characteristics of good software system that should be considered as essential for the system to be developed and justify your selection of each of them.
2. Why is the Waterfall model not appropriate for an Agile approach to software engineering?
3. Does Object-Oriented programming increase coupling and reduce cohesion? Explain.
4. Given the following class diagram, identify the main problem with it and indicate how it can be fixed.



Part III – Problem Solving Questions — 15 Marks

*Answer **all** the following problem-solving questions. Each question is worth 7.5 Marks.*

1. Develop a Use Case diagram for a POS system with the following requirements:
   a. The main actor is a user that can perform the tasks: create new bill, scan/record item, generate total.
   b. An administrator is a user that can perform all the above tasks in addition to: apply 10% discount, print user reports, check inventory.

   Important: Make sure to use Generalization properly and use stereotypes at least once.

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2. Develop an activity diagram for the task "Changing a registered course's section" in a university system assuming that the task can be completed by the student. Make sure to cover the following requirements:
   - Use Swimlanes to represent 2 roles, the Student and the RegistrationSystem.
   - The change of section can only be fulfilled if the section has remaining seats and the time does not conflict with another section that the student is already registered in.
   - Make sure to use at least 8 activities.

End of questions

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